Forces and mass

ALL MOTION IS CAUSED BY FORCES ACTING ON MASSES.

The effect of a force depends on the mass of the object. The greater the object's mass, the lower its resultant acceleration.

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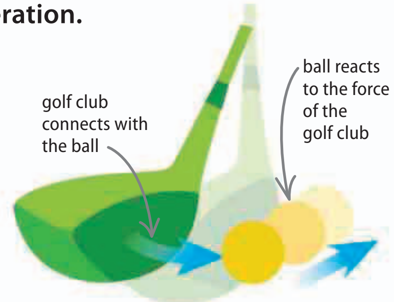
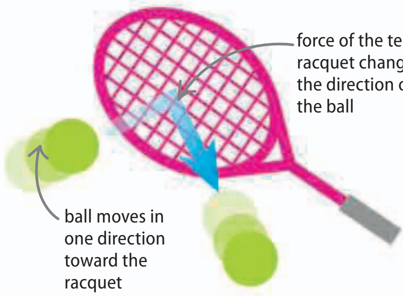
What is a force?

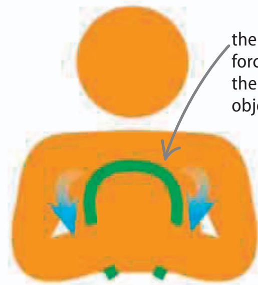
A force can affect an object in different ways. First, it may change the object's speed, so it moves faster or slower; second, a force can change the direction in which the object moves; third, the force may deform the shape of the object. Forces are measured in newtons (N). A force of 1 N results in a mass of 1 kilogram (kg) or 2.2 pounds (lb) reaching a speed of 1 meter (m) per second in one second.



◁ **Changing speed** The force of the golf club increases the speed of the ball from zero to a high speed, sending it down the golf course.



◁ **Changing direction** The force of the tennis racquet on the ball makes it stop traveling in one direction, and moves the ball in a new direction.

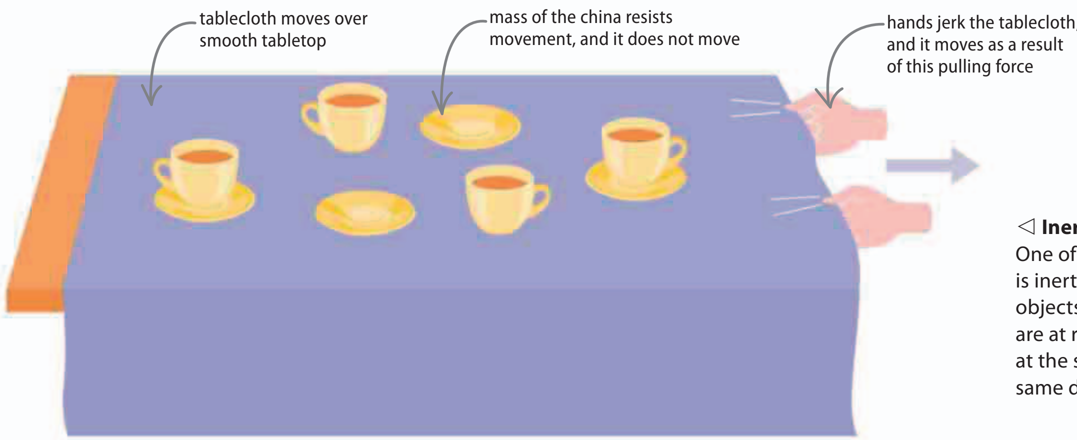


◁ **Changing shape** Depending on the toughness of the object, and strength of the person or machine used, the force exerted on an object may be able to change its shape.

What is mass?

Mass is a measure of how much an object resists a force. An object with a large mass contains more matter than one with a smaller mass. A force applied to a large mass results in a smaller acceleration than if it were applied to a small mass.

The precise kilogram unit is based on a single cylinder of the elements **platinum** and **iridium** kept in a safe in Paris, France.



◁ **Inertia** One of the properties of mass is inertia. This is a tendency for objects to remain where they are at rest—or keep traveling at the same speed and in the same direction if on the move.